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Predictive Modeling of Type 2 Diabetes Risk in U.S. Adults

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Abstract

Diabetes is one of the most common chronic diseases and was the eighth leading cause of death in the United States (US). The aim of this project is to create integrated and gender-specific predictive models to identify risk of an individual to develop type 2 diabetes, which could assist in early diagnosis and medication, reduce medical costs and disease prevention. This project used three widely used predictive modeling for binary target variable: decision tree, logistic regression, and neural network under different assessment measures. Neural Network was identified as best model for predicting diabetic. Both the logistic regression and Neural Network had high predictive power (81%). However, models' accuracy (74%) can still be improved. Age, Body Mass Index, Regular health checkup, difficulty in walking, heart disease, kidney disease, income, race, employment, physical health status, and blindness or difficulty seeing were the important factors identified by the model. People with increased age and excessive body weight have more likelihood to develop diabetes. The most critical age for diabetes checkups was identified as 35 to 49 years old. People who have difficulty walking or are blind and have difficulty seeing have also a higher likelihood. Moreover, people with low income, American Indian or Alaskan Native are at greater risk of developing diabetes.

1. Background

Diabetes[a] is one of the most prevalent chronic diseases and was the eighth leading cause of death in the United States in 2020[1]. Currently, 37.3 million Americans—11.3% of the US population—are living with diabetes (diagnosed and undiagnosed)[2]. People with diabetes can develop high blood pressure, high cholesterol, and high triglycerides (a type of fat in the blood). High blood sugar, particularly when combined with high blood pressure and high triglycerides or too much LDL ("bad") cholesterol levels can develop complications like heart disease, stroke, vision loss, kidney failure, lower-limb amputation, and even early death. The total number of diabetic prevalence cases has steadily increased in the USA since 2001[1]. Even though the number of newly diagnosed cases has decreased recently, the prevalence of prediabetes[b] among US adults has remained steady. According to the national diabetes report 2022, an estimated 96 million US adults—more than 1 in 3—have prediabetes, and more than 8 in 10 of them don't know they have it. A person with prediabetes is at high risk of type 2 diabetes, heart disease, and stroke [2]. Besides health threats, diabetes is also exerting a significant financial burden on the US economy. The country had a total estimated cost of $327 billion for diagnosed diabetes in 2017, including direct medical costs and reduced productivity[3]. According to American Diabetes Association, the

[a] **Diabetes** is a group of diseases characterized by high blood sugar. When a person has diabetes, the body either does not make enough insulin (**type 1**) or is unable to properly use insulin (**type 2**). When the body does not have enough insulin or cannot use it properly, blood sugar (glucose) builds up in the blood.

[b] **Prediabetes** is a serious health condition in which blood sugar levels are higher than normal, but not high enough yet to be diagnosed as type 2 diabetes.

average medical costs for people with diagnosed diabetes are 2.3 times higher than costs for people without diabetes.

While there is no cure for diabetes, strategies, like losing weight, eating healthily, being active, diagnose early, and receiving medical treatments, can mitigate the harms of this disease in many patients. Knowing the risk factors for diabetes is thus important for both prevention and starting early treatment of the disease. Type 1 diabetes is thought to be caused by an immune reaction, where the body attacks itself. Certain genes make people more likely to develop type 1 diabetes, and environmental factors may also play a part. This form of diabetes usually develops in children, teens, and young adults, but it can happen at any age. Currently, there is no known way to prevent type 1 diabetes, which accounts for approximately 5–10% of all diabetes cases[1]. Unlike type 1 diabetes, there are proven lifestyle changes (losing weight if overweight, eating healthier, and getting regular physical activity) that can reduce risk of developing type 2 diabetes[4]. Since 2012, the Centers for Disease Control and Prevention (CDC) has worked to prevent or delay type 2 diabetes by leading the National Diabetes Prevention Program (National DPP). The National DPP lifestyle change program is based on a randomized, controlled clinical trial study, conducted at 27 clinical centers around the United States from 1996 to 2001. After 3 years of study, the DPP showed that making realistic behavior changes helped people with prediabetes lose 5% to 7% of their body weight and reduce their risk of developing type 2 diabetes by 58% (71% for people older than 60)[5].

This study makes attempt to develop predictive modeling for diabetic based on Behavioral Risk Factor Surveillance System's (BRFFS) health-related survey data 2021. Predictive models for diabetes risk can be an important tool for the general public, public health officials, and public-private partners working to mitigate the effect of the disease. This will help to aware general public about the factors associated with diabetes and make them more concerned. Hence, this study has attempted to identify factors associated with type 2 diabetes in the US adult population, both aggregated and gender-specific.

2. Research Objective

The overall objective of the research is to develop a predictive model that can accurately predict an individual's risk of developing diabetes. The specific objectives of this research are as follows:

1. Develop a predictive model to identify the risk factors of type 2 diabetes among the US adult population.
2. Develop gender-specific predictive models to identify the risk factors of type 2 diabetes among US adult population.
3. To recommend the best predictive model for predicting diabetic cases.

3. Limitation of the study

1. Diabetic cases were self-reported; no medical test was undertaken to detect diabetes. So, there is a possibility of response bias in the diabetic data.
2. Some important factors of diabetes like 'sleep time' and 'controlled food habits' were missing in BRFSS survey 2021 data.

4. Literal Review

Article	Objective	Sample	Variables	Models	Results	Year	Limitations	Dataset
1. Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques[6]	Develop predictive models to identify risk factors for type 2 diabetes	138,146	279	Support vector machine, decision tree, logistic regression, random forest, neural network, and Gaussian Naive Bayes classifiers	Neural network model gave the best model performance (82.4%). people who slept 9 or more hours per day or had checkup frequency of less than 1 year had higher risk for type 2 diabetes.	2019	No prediction was made separately for male/female or by ethnicity/area.	BRFSS 2014 data
2. Predicting Risk of Type 2 Diabetes by Using Data on Easy-to-Measure Risk Factors	Determine the ability of models with nonlinear and interaction terms to assess the risk of type 2 diabetes.	5,471	-	multivariate adaptive regression splines (MARS)	-	2017	key risk factor for type 2 diabetes was increasing age, especially for those older than 69, followed by a family history of diabetes, with diminished risk among individuals younger than 45.	

What is different in this study? (relevance):

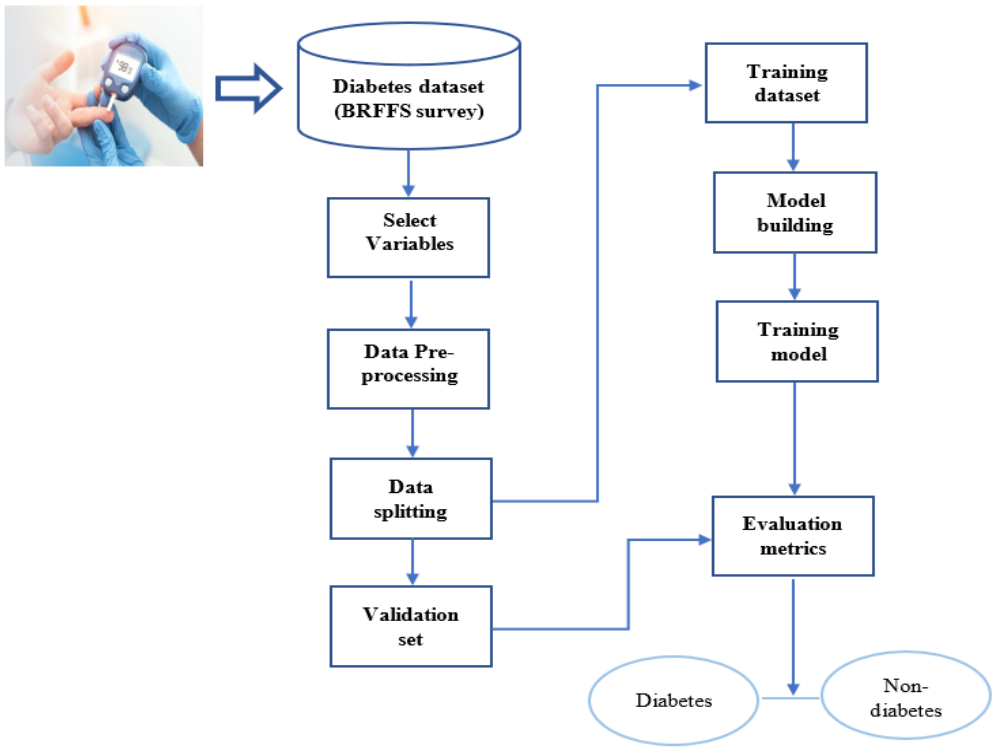
Many predictive models have been built previously, which identified risk factors for type 2 diabetes. However, this study attempts to use recent data from BRFSS to find the best predictive

model for diabetes prediction. Moreover, gender-specific predictive model was also generated to observe if they provide better accuracy.

5. Methodology

This section explains the methods used in the study, including data collection and attribute selection, building models, and extracting the required knowledge. This will follow a general order of preprocessing the data set, data cleaning, exploratory data analysis, checking for any violations in the multicollinearity assumption, building models with validation and testing, selecting a proposed model, checking any remaining assumptions, readjusting our final model, and drawing conclusions from the process as shown in Diagram 1.

Diagram 1: Methodology (workflow of the project)



5.1 Target and Inputs

Target: Diabetic cases (yes/no)

Inputs: State, depressive disorder, Age, Race, Sex, Marital status, Level of education, Income, Employment, Ownership of house, Physical activity, Routine checkup, Metropolitan status, Difficulty concentrating or remembering, Heart disease, Kidney disease, Difficulty in walking, Blind or difficulty seeing, Physical health status, Smoking status, Health insurance, Fruit Juice

intake, Fruit intake, Dark green vegetable intake, Potato servings, French fry intake, Other vegetable intake, Mental health status, Body mass index

Detail description of variables is attached in Appendix.

5.2 Data Processing

CDC conducts Behavioral Risk Factor Surveillance System (BRFSS) survey, which is a health-related telephone survey conducted annually from 1984. The combined landline and cellular telephone multiple version questionnaire data sets contain the data from the states which conducted more than one version of their landline and cellular telephone survey questionnaires and used optional modules in 2021. Each year about 400,000 individuals is surveyed, on average, to collect surveillance data on risk behaviors through monthly telephone interviews. We will use 2021 dataset for this project, which surveyed 438,693 individuals and covered all 50 states including the District of Columbia, Guam, Puerto Rico, and US Virgin Islands. All the relevant variables from the record of 229 variables will be used for model building. All the responses in the survey were self-reported and no proxy interviews were taken. Dataset is available on CDC's 2021 BRFSS Survey Data and Documentation page.

BRFSS data is available in American Standard Code for Information Interchange (ASCII) format. We used SPSS to import data from ASCII format and initial data preparation. The data preparation task includes identifying necessary variables and excluding irrelevant cases. Initially, we had 438693 observations. However, we excluded those who are under 18 years old, had prediabetic, and pregnant women from our analysis. So, finally, we had 423958 observations for model building. After data preparation, we imported data into SAS Miner Enterprise. We used SAS Miner Enterprise 15.2 for final data cleaning and modeling. "Metropolitan status" (city/county/suburban) was excluded due to a large number of missing values in the dataset and "State code" was excluded due to a large number of categories. Following is the variable summary for our modeling after importing data in SAS Miner Enterprise:

Variable Summary

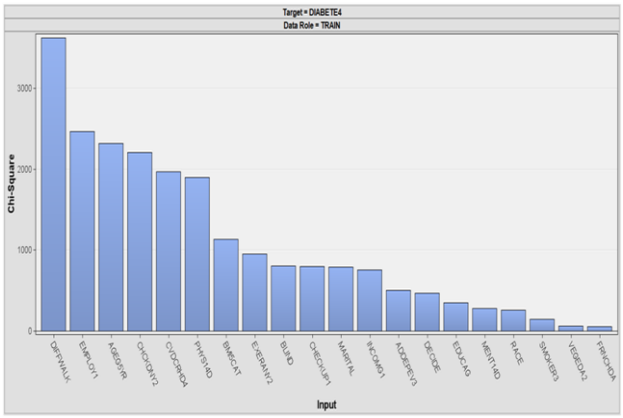
Role	Measurement Level	Frequency Count
ID	INTERVAL	1
INPUT	BINARY	10
INPUT	NOMINAL	17
REJECTED	NOMINAL	2
TARGET	BINARY	1

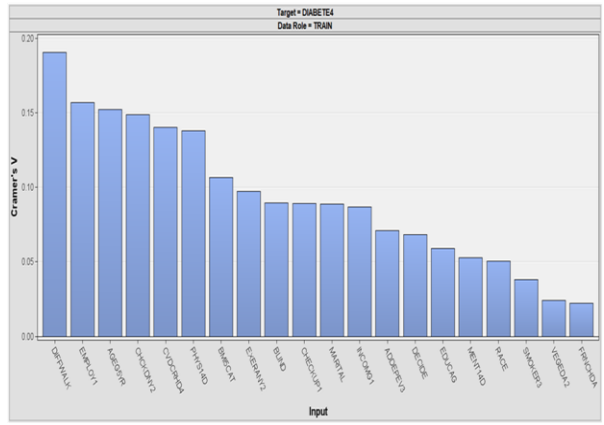
5.3 Exploratory Data Analysis

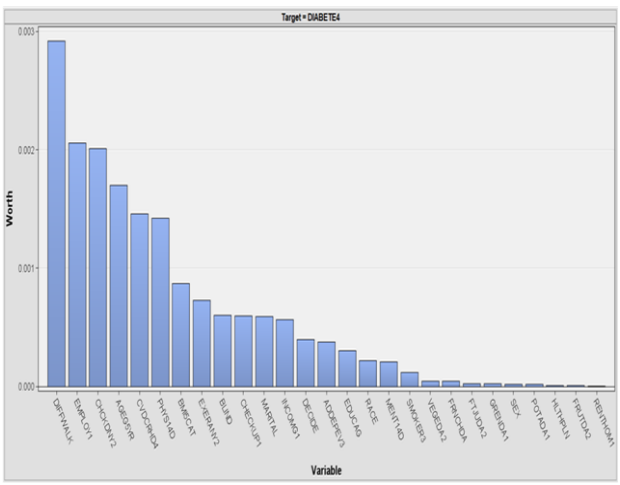
Before model building, we performed exploratory data analysis to identify variables with missing values and outliers, minimizing noises, cleaning data, attribute aggregations, discretization and binarization, attribute transformation, reparameterization etc.

From exploratory data analysis, we see that our target variable is imbalanced (Nondiabetic case: 86.4% vs Diabetic case: 13.6%). So, we decided to balance the data before modeling.

Following are the importance and worth of variables for predicting diabetic based on Chi-square, Cramer's V value. We included all the important variables in our modeling.







5.4 Model Building

Our model building phase started by checking the balance of the data set. As Diabetic data was imbalanced (Diabetic: 13.6% and Non-diabetic: 86.4%), we balanced data into 50:50 ratio using sample node in SAS Miner enterprise.

Data=DATA

Before balancing data

Variable	Numeric Value	Formatted Value	Frequency Count	Percent	Label
DIABETE4	0	0	366342	86.4100	DIABETE4
DIABETE4	1	1	57616	13.5900	DIABETE4

Data=SAMPLE

After balancing data

Variable	Numeric Value	Formatted Value	Frequency Count	Percent	Label
DIABETE4	0	0	57616	50	DIABETE4
DIABETE4	1	1	57616	50	DIABETE4

Three commonly used prediction models for binary target data were used for building the prediction modeling:

1. Logistic Regression
2. Decision Tree
3. Neural Network

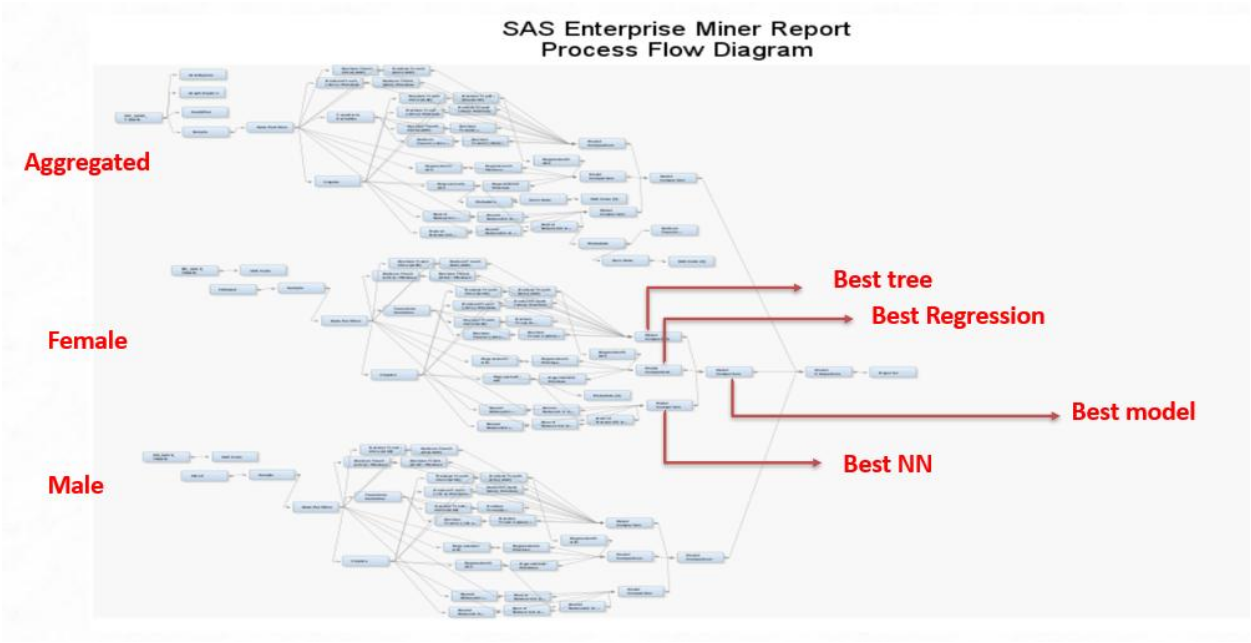
Above models are well recognized for modeling target binary variable. We used 50:30:20 ratio for training, validation and testing data.

Partition Summary

Type	Data Set	Number of Observations
DATA	EMWS1.Smpl_DATA	115232
TRAIN	EMWS1.Part2_TRAIN	57615
VALIDATE	EMWS1.Part2_VALIDATE	34570
TEST	EMWS1.Part2_TEST	23047

For modeling purpose, we performed 12 different decision tree, 5 logistic regression and 4 neural network models based on different assessment measure. The best of decision tree, logistic regression model and neural network used to choose the best model for diabetic predictive modeling (Diagram 2).

Diagram 2: Process Flow Diagram



Model Performance Measure

In each category method, assessment metrics are important in defining whether the method is successful in learning of the data. Consequently, to establish a model's performing, matrix is employed to assess the amount of False Positive (F.P.), True Negatives (T.N.), False Negatives (F.N.), and True Positives (T.P.). Furthermore, the actual results were assessed to the outcomes of the categorization model as following:

- ✓ T.P.: The model predicts the patients will have diabetes, and the patients had a diabetes
- ✓ TN: The model predicts the patients will not have diabetes, and the patients did not have diabetes.
- ✓ F.P.: The model predicts the patients will have diabetes, and the patients did not have a diabetes.
- ✓ F.N.: The model predicts the patients will not have diabetes, and the patients have diabetes.

This project will performance model as following:

- Model accuracy

The accuracy method was applied to measure the model proficiency, as taken in Equation (1).

$$\text{Accuracy} = \frac{TP+TN}{TP+\text{FN}+\text{TN}+FP} \quad (1)$$

- Model Recall (Sensitivity of the model)

The recall is the proportion among the total of positive rates predicted and the total of positive rates with false negative that should be calculated.

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

- Model $F - Score$

F1 score is an assessment of the accuracy of the model. To calculate the score both precision and recalls are calculated as below:

$$F1 - Score = \frac{2*Recall*\text{accuracy}}{\text{accuracy} + Recall} \quad (3)$$

F1-score can range from 0-1. A higher F1- score means that the model is better in terms of both precision and recalls.

- ROC curves and AUC

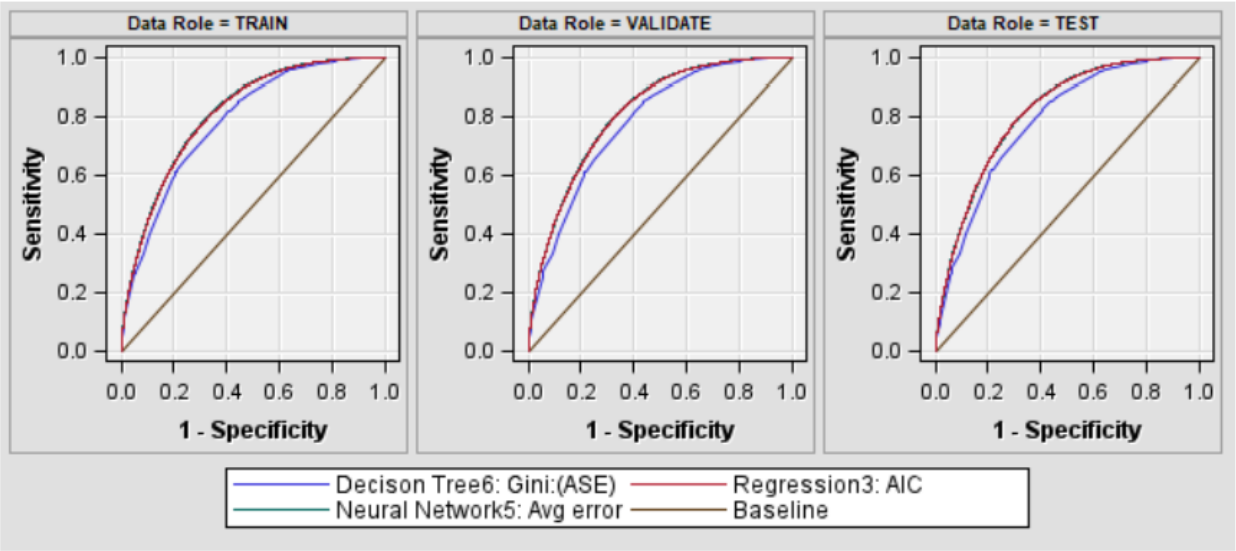
To assess the model performance, the area under the curve (AUC) and receiver Operating Characteristics curve (ROC) are considered. Where rates of AUC change over between 0 and 1 as well as classes. The ROC is a graphic demonstration the "TP," and "FP" in categorization effects of the ML method. ROC curve demonstrates the models' predictive power of the model. The higher the ROC (ranges from 0-1), the higher the model's predictive power. For a model to have at least a minimum predictive power, the ROC index should be greater than 0.5.

6. Results and Discussion

6.1 Best Predictive Model

We see from the following ROC index that both logistic regression model and neural network have higher ROC value i.e. these two models performed better than the decision tree model. However, there was no significant difference between logistic regression model and neural network model based on ROC.

Figure 1: ROC index for the best three predictive models



For logistic regression model, the ROC value was 0.810, while for neural network it was 0.811. On the other hand, the ROC index for decision tree model was 0.779. F1-score was also slightly better for neural network model: 0.76 for neural network compared to 0.75 for logistic regression model (Table 1). The accuracy of these two models were same (74%). Based on these values, we can say that Neural network model was the best model. However, the performance of the neural network and regression model was almost same.

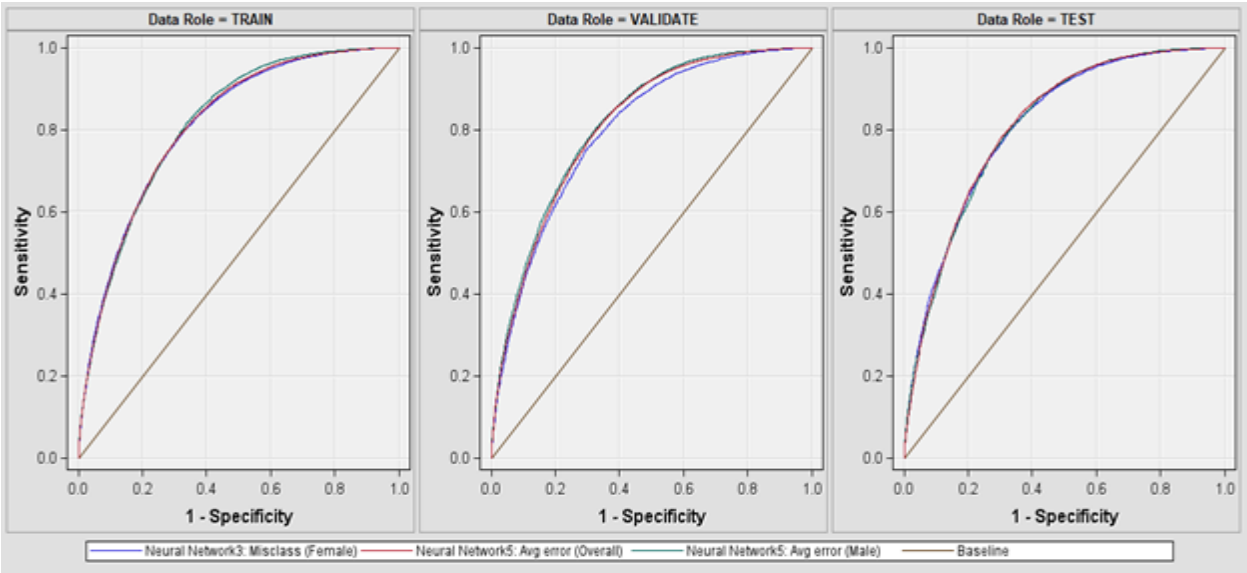
Table 1: Models performance measure

Model	Accuracy	Average Squared Error	Cumulative Lift (at Depth 20)	ROC Index	F1-score
Logistic regression	0.74	0.18	1.74	0.810	0.75
Decision Tree	0.70	0.19	1.66	0.779	0.67
Neural Network	0.74	0.18	1.73	0.811	0.76

6.2 Gender-Specific Predictive Models

As we see from Table 1, model accuracy was not as expected. So, we performed gender-specific models to see if the accuracy of the model improves. Figure 2 depicts ROC index comparison between the best modes from aggregated model and from gender-specific models (female and male). It is clear from the figure that gender-specific models do not perform better than the aggregated one. So, we used aggregated model for interpretation of results.

Figure 2: ROC index of the best models from aggregated and gender-specific models



6.3 Important Factors of Diabetes

Predicted target of Neural Network model was applied to Decision Tree to identify the importance of the factors. Since both the Neural Network and Logistic regression models performed almost equally well and all the important variables in the Neural network are also significant in regression model, the odds ratio from the regression model was used to interpret outputs for simplicity.

Table 1 presents important factors identified by the model. According to our analysis, age, body mass index, regular health checkup and difficulty walking are the most important factors. Heart disease and Kidney disease, Income, race, employment, physical health status, blindness have also significant impact on diabetic.

Table 2: Importance of Diabetic Factors

Factors	Importance
Age	1.000
BMI (Body Mass Index)	0.7238
Regular health checkup	0.4779
Difficulty walking	0.4753
Heart disease	0.2043
Kidney disease	0.1715
Income	0.1704
Race	0.1067
Employment	0.0657
Physical health status	0.0461
Blindness or difficulty seeing	0.0314

Age: Age is the most important factor for diabetes, as identified by the model. Age 80+ was the reference category for estimating odds ratio. As we can see, with age the odds ratio of diabetic are increasing. We calculated increased risk (in %) of diabetes from one age group to the next age group. Age 35-59 was the critical time point when people have higher risk of developing diabetic. Age 35-39 was identified as the most critical age group when the likelihood of developing diabetic increases by 61 percent from its previous age group (age 30-34).

Table 3: Impact of age on diabetes development

Age	Odds Ratio	% Increased Risk
Age 18 to 24	0.07	-
Age 25 to 29	0.09	23
Age 30 to 34	0.11	25
Age 35 to 39	0.18	61
Age 40 to 44	0.28	55
Age 45 to 49	0.41	44
Age 50 to 54	0.54	33

Age	Odds Ratio	% Increased Risk
Age 55 to 59	0.73	34
Age 60 to 64	0.84	16
Age 65 to 69	0.91	8
Age 70 to 74	1.08	19
Age 75 to 79	1.05	-3
Age 80+	**Reference category**	

Body Mass Index (BMI): Having excessive body weight increases the risk of developing diabetes. We found that obese people are at 6 times higher risk to develop diabetes compared to those underweight, 3 times higher risk compared to those normal weight, and 2 times higher risk compared to those overweight.

Regular health checkup: It is expected that with regular health checkup, likelihood of diabetes will decrease. However, according to the odds ratio in Table 4, we see that people who performed regular health checkup in less than 2 years are more likely to develop diabetes than people who never performed regular health checkup. However, the next two groups, people who performed regular health checkup within 3-5 years or more years are more likely to develop diabetes than people who never performed regular health checkup. Underlying reason for this result can be that the diabetes group themselves have contributed to the result. BRFSS survey did not collect this data retrospectively i.e. people's practice before developing diabetes, rather they asked their current practice on the day of survey. So, it is more likely that people who are already suffering from diabetes needs to have a frequent health checkup.

Table 4: Impact of regular health check up on diabetes

Last health checkup performed	Odds ratio
<1 year	2.69
1-2 years	1.19
3-5 years	0.87

5+	0.68
Never	**Reference category**

Difficulty Walking: People who have difficulty walking or climbing stairs have a higher risk of developing diabetes: people who have difficulty walking or climbing stairs have 33 percent more chance to be affected by diabetes.

Coronary Heart Disease: People affected by coronary heart disease are more likely to have diabetes. The estimated odds ratio for people with coronary heart disease to those without coronary heart disease in 1.972.

Kidney Disease: People affected by kidney disease have more likelihood to have diabetes. The estimated odds ratio for people with kidney disease to those without kidney disease is 2.579.

Income: Low-income people are more likely to develop diabetes than high-income people.

Race: American Indians or Alaskan Natives are at a greater risk of developing diabetes, followed by Asians.

Employment: People unable to work or unemployed are more likely to develop diabetes than people who are employed or self-employed.

Physical Health Status: People with good physical health status have less risk to develop diabetes.

Blindness of Difficulty Seeing: People with blindness are 35 percent more likely to develop diabetes.

7. Conclusion and Recommendations

The Neural Network is the best prediction model for diabetes identified by this study. Both Neural Network and Logistic regression model performed well in terms of predictive power (81%). However, prediction accuracy was not as expected. The accuracy of the model can be improved by adding other possibly important inputs like ‘sleep time’ and ‘controlled food habits’ is the model. BRFSS survey 2021 does not include the relevant data. So, further research can be done by including these important factors in the model.

From the study results, age and body mass index were the top two factors of diabetes. People entering the age of 35-39 age group should take steps like controlling weight and performing regular health checkups to prevent or delay diabetes development. Public-private partners working to mitigate the effect of the disease should focus on low-income people and people with disabilities or impairments such as blindness or difficulty in walking or climbing stairs to mitigate the effect.

8. References

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9. Appendix

Target and input variables selected from BRFSS 2021 survey:

Indicators	Role	Name	Type	Code
Have Diabetes?	Target	DIABETE4	Binary	1=Yes 2=No Note: Respondents younger than 18 years (possible type 1 diabetes), who were pregnant, and respondents with prediabetes were excluded.
State	Input	STATE	Categorical	51 states of the USA
Have Depressive Disorder?	Input	ADDEPEV3	Binary	1=Yes 2=No
Age	Input	AGEG5YR	Categorical	1=Age 18 to 24 2=Age 25 to 29 3=Age 30 to 34 4=Age 35 to 39 5=Age 40 to 44 6=Age 45 to 49 7=Age 50 to 54 8=Age 55 to 59 9=Age 60 to 64 10=Age 65 to 69 11=Age 65 to 69 12=Age 65 to 69 13=Age 75 to 79
Race	Input	RACE	Categorical	1=White 2=Black 3=American Indian or Alaskan Native 4=Asian 5=Native Hawaiian or other Pacific Islander 6=Other race 7=Multiracial
Sex	Input	SEX	Categorical	1=Male 2=Female
Marital Status	Input	MARITAL	Categorical	1=Married 2=Divorced 3=Widowed 4=Seperated 5=Never married

				6=A member of an unmarried couple
Level Of Education	Input	EDUCAG	Categorical	1=Did not graduate High School 2=Graduated High School 3=Attended College or Technical School 4=Graduated from College or Technical School
Income	Input	INCOMG1	Categorical	1=Less than $15,000 2=$15,000 to less than $25,000 3=$25,000 to less than $35,000 4=$35,000 to less than $50,000 5=$50,000 to less than $100,000 6=$100,000 to less than $200,000 7=$100,000 to less than $200,000
Employment	Input	EMPLOY1	Categorical	1=Employed 2=Self-employed 3=No work >1 y 4=No work <1 y 5=Homemaker 6=Student 7=Retired 8=Unable to work
Ownership Of House	Input	RENTHOM1	Categorical	1=Own 2=Rent 3=Other
Physical Activity	Input	EXERANY2	Categorical	1=yes 2=no
Routine Checkup	Input	CHECKUP1	Categorical	1=1 years 2=1-2 years 3=3-5 years 4= 5+ years 5=Never
Metropolitan Status	Input	MSCODE	Categorical	1=Center city 2=County 3=Suburban 4=not in MSA
Difficulty Concentrating Or Remembering	Input	DECIDE	Binary	1=yes 2=no
Heart Disease	Input	CVDCRHD4	Binary	1=yes

				2=no
Have Kidney Disease?	Input	CHCKDNY2	Categorical	1=yes 2=no
Difficulty In Walking?	Input	DIFFWALK	Categorical	1=yes 2=no
Blind Or Difficulty Seeing	Input	BLIND	Binary	1=Yes 2=No
Physical Health Status	Input	PHYS14D	Categorical	1=0 days physical health not good 2=1-13 days physical health not good 3=14+ days physical health not good
Smoking Status	Input	SMOKER3	Categorical	1=Current smoker–now smokes every day 2=Current smoker–now smokes some days 3=Current smoker–now smokes some days 4=Never smoked
Have Any Health Insurance?	Input	HLTHPLN	Binary	1= Yes 2= No
Fruit Juice Intake	Input	FTJUDA2	Categorical	1=daily/weekly 2=monthly 3=never
Fruit Intake	Input	FRUTDA2	Categorical	1=daily/weekly 2=monthly 3=never
Dark Green Vegetable Intake	Input	GRENDA1	Categorical	1=daily/weekly 2=monthly 3=never
Potato Servings	Input	POTADA1	Categorical	1=daily/weekly 2=monthly 3=never
French Fry Intake	Input	FRNCHDA	Categorical	1=daily/weekly 2=monthly 3=never
Other Vegetable Intake	Input	VEGEDA2	Categorical	1=daily/weekly 2=monthly 3=never

Mental Health Status	Input	MENT14D		1=0 days physical health not good 2=1-13 days physical health not good 3=14+ days physical health not good
Body Mass Index	Input	BMI5CAT		1=Underweight 2=Normal weight 3=Overweight 4=Obese